

FINAL BUSINESS INSIGHTS REPORT

E-Commerce Data Analytics Internship Project

Organization	Primeor Solutions
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Tools	Excel • SQL • Power BI
Dashboard scope	Executive Overview • Sales Analysis • Product & Customer Insights

1. Executive Summary

The E-Commerce dataset covers 51,258 order lines placed between 2011 and 2014 across seven markets and thirteen regions. The business generated ₹12.63M in total sales and ₹1.47M in net profit, an overall profit margin of 11.6%. The dashboard suite built for this project (Executive Overview, Sales Analysis, and Product & Customer Insights pages) surfaces the following headline metrics:

TOTAL SALES ₹12.63M	TOTAL PROFIT ₹1.47M	TOTAL ORDERS 51.26K	AVG. DISCOUNT 14.29%
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Sales grew every year of the observed period – from ₹2.26M in 2011 to ₹4.29M in 2014, an 90% increase – while nearly a quarter of all order lines (24.5%) were sold at a loss, pointing to a discounting and product-mix problem that outweighs pure demand growth.

2. Sales Performance Overview

2.1 Regional & Market Distribution

The Central region is the single largest contributor, generating ₹2.82M – more than 1.5x the next-highest region (South, ₹1.60M). At the market level, APAC leads with ₹3.58M (28% of total sales), followed by EU (₹2.94M) and US (₹2.30M). Canada is the smallest market by a wide margin, contributing under ₹0.07M.

Region	Total Sales
Central	₹2.82M
South	₹1.60M
North	₹1.25M
Oceania	₹1.10M
Southeast Asia	₹0.88M
North Asia	₹0.85M

2.2 Monthly & Yearly Trend

Monthly sales are highly seasonal: every year shows a trough in February and a sharp run-up into November–December, consistent with year-end / holiday purchasing. The strongest month recorded was December at ₹1.58M, more than double the weakest month, February (₹0.54M). Year-over-year, sales rose consistently: ₹2.26M (2011) → ₹2.68M (2012) → ₹3.40M (2013) → ₹4.29M (2014), a compound growth pattern that suggests the business is scaling well but also becoming more dependent on Q4 performance.

- Recommendation: build inventory and staffing plans around the Nov–Dec peak, and design a Feb–Apr promotional push to smooth the seasonal trough.

3. Category & Sub-Category Analysis

Technology is the strongest category on every dimension that matters: it leads in sales (₹4.74M) and by far in average profit per order (₹65.4 vs. ₹29.0 for Furniture and ₹16.6 for Office Supplies), while carrying the lowest average discount (13.5%). Furniture, despite being the second-largest category by sales (₹4.11M), is the least profitable per order and is discounted the most heavily (16.8% on average) – a combination that erodes margin.

3.1 Profit by Sub-Category

Copiers, Phones and Bookcases are the most profitable sub-categories, together contributing over ₹636K in profit. Tables is the only sub-category with a net loss across the full dataset, at –₹64.1K, and Fasteners and Labels are barely profitable. This mirrors the dashboard's "Profit by Sub-Category" chart, where the bars taper from Copiers down to a negative bar for Tables.

Sub-Category	Total Profit
Copiers	₹258.2K
Phones	₹216.7K
Bookcases	₹161.7K
Appliances	₹141.7K
Chairs	₹141.7K
Tables	–₹64.1K

- Recommendation: review Tables pricing and discount policy specifically – it is the only sub-category losing money overall, likely driven by high shipping cost and excessive discounting on bulky items.

4. Discounting & Profitability Risk

Across the full dataset, 12,538 order lines (24.5% of all orders) were sold at a net loss, totalling –₹920K in negative profit. Furniture's above-average discount rate (16.8%) combined with its low per-order profit indicates that discount depth, not just category mix, is a key driver of the loss-making orders. Because the average discount across the business is already 14.3%, any further blanket discounting is likely to push more order lines into loss territory rather than grow volume profitably.

- Recommendation: cap discounts on Furniture and Office Supplies sub-categories with historically thin or negative margins, and introduce discount-approval thresholds above ~15%.

5. Product & Customer Insights

5.1 Top-Selling Products

Smartphones dominate the top of the product list – Apple, Cisco, Motorola and Nokia smart phones occupy four of the top five positions by sales, led by the Apple Smart Phone, Full Size at ₹87K. Executive leather armchairs (Hon, Office Star, Harbour Creations) form the next cluster, confirming Technology and premium Furniture as the two hero product groups.

5.2 Top Customers

Sales are not overly concentrated in a handful of accounts: the top customer, Tom Ashbrook, accounts for ₹40K (well under 1% of total sales), and the top 10 customers combined are a small fraction of the ₹12.63M total – a healthy sign that revenue is broad-based rather than dependent on a few key accounts.

5.3 Segment Mix

The Consumer segment is the primary revenue driver at ₹6.50M (51% of sales), followed by Corporate at ₹3.82M (30%) and Home Office at ₹2.31M (18%). This ranking matches the dashboard's segment donut chart and suggests marketing spend aimed at individual consumers currently yields the highest volume return.

6. Operations: Shipping & Fulfilment

Standard Class is the dominant shipping mode, used in 30,764 of 51,271 orders (60%), followed by Second Class (10,304), First Class (7,504) and Same Day (2,699). The average gap between order date and ship date across the dataset is about 4 days. The heavy reliance on Standard Class keeps logistics costs down but may be worth testing against faster options for the High/Critical priority orders, which together make up over 38% of all orders.

7. Key Recommendations

- Protect margin on Furniture: tighten discount limits and renegotiate supplier costs on Tables, the only loss-making sub-category.
- Double down on Technology: it delivers the highest profit per order at the lowest discount – prioritise it in promotions and inventory planning.
- Plan for seasonality: pre-build stock and staffing for the Nov–Dec peak; run targeted campaigns in Feb–Apr to lift the seasonal trough.
- Introduce a discount governance rule: flag or require approval for discounts above ~15%, since average discount already sits at 14.3% and a quarter of orders are unprofitable.
- Grow the Home Office segment: it is currently the smallest of the three segments (18% of sales) and represents headroom for targeted marketing.
- Review APAC and EU logistics: as the two largest markets, small improvements in shipping cost or delivery speed there will have the largest absolute impact.

8. Conclusion

The Power BI dashboards built for this project (Executive Overview, Sales Analysis, and Product & Customer Insights) confirm a business that is growing steadily –year-over-year sales nearly doubled from 2011 to 2014– but whose profitability is being held back by aggressive discounting in specific categories, most notably Furniture and its Tables sub-category. Technology and the Consumer segment are the clearest strengths to build on, while a tighter discount policy and seasonal planning represent the fastest levers to improve overall margin.